9. Because both the given polynomials are irreducible over $\mathbb{Z}_2$, both $\mathbb{Z}_2(\alpha)$ and $\mathbb{Z}_2(\beta)$ are extension of $\mathbb{Z}_2$ of degree 3 and thus are subfields of $\overline{\mathbb{Z}}_2$ containing $2^3 = 8$ elements. By Theorem 33.3, both of these fields must consist precisely of the zeros in $\overline{\mathbb{Z}}_2$ of the polynomial $x^8 - x$. Thus the fields are the same.

10. Let $p(x)$ be irreducible of degree m in $\mathbb{Z}_p[x]$. Let K be the finite extension of $\mathbb{Z}_p$ obtained by adjoining *all* the zeros of $p(x)$ in $\overline{\mathbb{Z}}_p$. Then K is a finite field of order p^n for some positive integer n, and consists precisely of all zeros of $x^{p^n} - x$ in $\overline{\mathbb{Z}}_p$. Now $p(x)$ factors into linear factors in $K[x]$, and these linear factors are among the linear factors of $x^{p^n} - x$ in $K[x]$. Thus $p(x)$ is a divisor of $x^{p^n} - x$.

11. Because $\alpha \in F$, we have $\mathbb{Z}_p(\alpha) \subseteq F$. But because α is a generator of the multiplicative group F^*, we see that $\mathbb{Z}_p(\alpha) = F$. Because $|F| = p^n$, the degree of α over $\mathbb{Z}_p$ must be n.

12. Let F be a finite field of p^n elements containing (up to isomorphism) the prime field $\mathbb{Z}_p$. Let m be a divisor of n, so that $n = mq$. Let $\overline{F} = \overline{\mathbb{Z}}_p$ be an algebraic closure of F. If $\alpha \in \overline{\mathbb{Z}}_p$ and $\alpha^{p^m} = \alpha$, then $\alpha^{p^n} = \alpha^{p^{mq}} = (\alpha^{p^m})^{p^{m(q-1)}} = \alpha^{p^{m(q-1)}} = (\alpha^{p^m})^{p^{m(q-2)}} = \alpha^{p^{m(q-2)}} = \cdots = \alpha^{p^m} = \alpha$. By Theorem 33.3, the zeros of $x^{p^m} - x$ in $\overline{\mathbb{Z}}_p$ form the *unique* subfield of $\overline{\mathbb{Z}}_p$ of order p^m. Our computation shows that the elements in this subfield are also zeros of $x^{p^n} - x$, and consequently all lie in the field F, which by Theorem 33.3 consists of all zeros of $x^{p^n} - x$ in $\overline{\mathbb{Z}}_p$.

13. Let F be the extension of $\mathbb{Z}_p$ of degree n, consisting of all zeros of $x^{p^n} - x$ by Theorem 33.3. Each $\alpha \in F$ is algebraic over $\mathbb{Z}_p$ and has degree that divides n by Theorem 31.4. Thus each $\alpha \in F$ is a zero of a monic irreducible polynomial of a degree dividing n. Conversely, a zero β of an irreducible monic polynomial having degree m dividing n lies in a field $\mathbb{Z}_p(\beta)$ of p^m elements that is contained in F by Exercise 12. Thus the elements of F are precisely the zeros of all monic irreducible polynomials in $\mathbb{Z}_p[x]$ of degree dividing n, as well as precisely all zeros of $x^{p^n} - x$. Factoring into linear factors in $F[x]$, we see that both $x^{p^n} - x$ and the product g(x) of all monic polynomials in $\mathbb{Z}_p[x]$ of degree d dividing n have the factorization $\prod_{\alpha \in F}(x - \alpha)$, so $x^{p^n} - x = g(x)$.

14. **a.** Now $x^2 \equiv a \pmod{p}$ has a solution in $\mathbb{Z}$ if and only if $x^2 = b$ has a zero in $\mathbb{Z}_p$ where b is the remainder of a modulo p. Now $\mathbb{Z}_p^*$ is cyclic of order $p - 1$. The elements b of a cyclic group that are squares are those that are even powers of a generator, and these are precisely the elements b satisfying $b^{(p-1)/2} = 1$. Thus we see that $x^2 \equiv a \pmod{p}$, for a not congruent to zero modulo p, has a solution in $\mathbb{Z}$ if and only if $a^{(p-1)/2} \equiv 1 \pmod{p}$.

b. We know that $x^2 - 6$ is irreducible in $\mathbb{Z}_{17}[x]$ if and only if it has no zero in $\mathbb{Z}_{17}$, so that $6 \neq b^2$ for any $b \in \mathbb{Z}_{17}$. By Part(**a**), we test by computing whether $6^{(17-1)/2} = 6^8$ is congruent to 1 modulo 17. Computing in $\mathbb{Z}_{17}$, we have $6^2 = 2, 6^4 = 2^2 = 4$, and $6^8 = 4^4 = 16$, so 6 is not a square in $\mathbb{Z}_{17}$ and $x^2 - 6$ is irreducible.

34. Isomorphism Theorems

1. (See the answer in the text.)

2. **a.** $K = \{0, 6, 12\}$

b.
$$0 + K = \{0, 6, 12\}, \quad 1 + K = \{1, 7, 13\},$$
$$2 + K = \{2, 8, 14\}, \quad 3 + K = \{3, 9, 15\},$$
$$4 + K = \{4, 10, 16\}, \quad 5 + K = \{5, 11, 17\}$$

c. $\phi[\mathbb{Z}_{18}]$ is the subgroup $\{0, 2, 4, 6, 8, 10\}$ of $\mathbb{Z}_{12}$.

d. $\mu(0+K)=0, \mu(1+K)=10, \mu(2+K)=8, \mu(3+K)=6, \mu(4+K)=4, \mu(5+K)=2$

3. (See the text answer.)

4. a. $HN=\{0,3,6,9,12,15,18,21,24,27,30,33\}$

b. $0+N=\{0,9,18,27\},\ \ 3+N=\{3,12,21,30\},\ \ 6+N=\{6,15,24,33\}$

c. $0+(H\cap N)=\{0,18\}, 6+(H\cap N)=\{6,24\}, 12+(H\cap N)=\{12,30\}$

d. $\phi(0+N)=0+(H\cap N), \phi(3+N)=12+(H\cap N), \phi(6+N)=6+(H\cap N)$

5. (See the text answer.)

6. a.

$0+H=\{0,9,18,27\},\ 1+H=\{1,10,19,28\}$
$2+H=\{2,11,20,29\},\ 3+H=\{3,12,21,30\}$
$4+H=\{4,13,22,31\},\ 5+H=\{5,14,23,32\}$
$6+H=\{6,15,24,33\},\ 7+H=\{7,16,25,34\}$
$8+H=\{8,17,26,35\}$

b.

$0+K=\{0,18\},\ 1+K=\{1,19\},\ 2+K=\{2,20\}$
$3+K=\{3,21\},\ 4+K=\{4,22\},\ 5+K=\{5,23\}$
$6+K=\{6,24\},\ 7+K=\{7,25\},\ 8+K=\{8,26\}$
$9+K=\{9,27\},\ 10+K=\{10,28\},\ 11+K=\{11,29\}$
$12+K=\{12,30\},\ 13+K=\{13,31\},\ 14+K=\{14,32\}$
$15+K=\{15,33\},\ 16+K=\{16,34\},\ 17+K=\{17,35\}$

c. $0+K=\{0,18\},\ 9+K=\{9,27\}$

d.

$(0+K)+H/K=\{0+K,9+K\}=\{\{0,18\},\{9,27\}\}$
$(1+K)+H/K=\{1+K,10+K\}=\{\{1,19\},\{10,28\}\}$
$(2+K)+H/K=\{2+K,11+K\}=\{\{2,20\},\{11,29\}\}$
$(3+K)+H/K=\{3+K,12+K\}=\{\{3,21\},\{12,30\}\}$
$(4+K)+H/K=\{4+K,13+K\}=\{\{4,22\},\{13,31\}\}$
$(5+K)+H/K=\{5+K,14+K\}=\{\{5,23\},\{14,32\}\}$
$(6+K)+H/K=\{6+K,15+K\}=\{\{6,24\},\{15,33\}\}$
$(7+K)+H/K=\{7+K,16+K\}=\{\{7,25\},\{16,34\}\}$
$(8+K)+H/K=\{8+K,17+K\}=\{\{8,26\},\{17,35\}\}$

e.

$\phi(0+H)=(0+K)+H/K,\ \ \phi(1+H)=(1+K)+H/K$
$\phi(2+H)=(2+K)+H/K,\ \ \phi(3+H)=(3+K)+H/K$
$\phi(4+H)=(4+K)+H/K,\ \ \phi(5+H)=(5+K)+H/K$
$\phi(6+H)=(6+K)+H/K,\ \ \phi(7+H)=(7+K)+H/K$
$\phi(8+H)=(8+K)+H/K$

7. Let $x\in H\cap N$ and let $h\in H$. Because $x\in H$ and H is a subgroup, we know that $hxh^{-1}\in H$. Because $x\in N$ and N is normal in G, we also know that $hxh^{-1}\in N$. Thus $hxh^{-1}\in H\cap N$, so $H\cap N$ is a normal subgroup of H.

8. a. Let $\gamma: G\to G/H$ be the natural homomorphism of a group onto its factor group. Then $\gamma[K]=K/H=B$ is a normal subgroup of $A=G/H$ by Theorem 15.16. Similarly $\gamma[L]=L/H=C$ is a